Light

GCSE Computer Science - Fetch-Decode-Execute Cycle and Embedded Systems (Set 1)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What happens during the 'fetch' stage of the fetch-decode-execute cycle?

- ○ **A.** The next instruction is retrieved from main memory
- ○ **B.** The instruction is carried out by the CPU
- ○ **C.** The instruction is translated into machine code
- ○ **D.** The result is stored in secondary storage

Q2 (1 mark)

What is the purpose of the 'decode' stage in the fetch-decode-execute cycle?

- ○ **A.** To compress the instruction to save memory
- ○ **B.** To copy the instruction from storage to RAM
- ○ **C.** To display the instruction on the screen
- ○ **D.** To interpret the instruction and determine what action is required

Q3 (1 mark)

What occurs during the 'execute' stage of the fetch-decode-execute cycle?

- ○ **A.** The instruction is retrieved from memory
- ○ **B.** The instruction is interpreted by the control unit
- ○ **C.** The instruction is carried out, which may involve calculations or memory access
- ○ **D.** The instruction is deleted from the system

Q4 (1 mark)

What is the main function of the Arithmetic Logic Unit (ALU)?

- ○ **A.** To perform calculations and logical comparisons
- ○ **B.** To store programs permanently
- ○ **C.** To connect the computer to the internet
- ○ **D.** To display graphics on the monitor

Q5 (1 mark)

What is the role of the control unit in a CPU?

- ○ **A.** To perform mathematical calculations
- ○ **B.** To store data temporarily during processing
- ○ **C.** To provide power to the processor
- ○ **D.** To coordinate and manage the operations of all CPU components

Q6 (1 mark)

What is the purpose of the clock in a CPU?

- ○ **A.** To display the current time to the user
- ○ **B.** To synchronise the operations of all CPU components with regular pulses
- ○ **C.** To cool down the processor during intensive tasks
- ○ **D.** To connect to atomic time servers

Q7 (1 mark)

What is a 'bus' in a computer system?

- ○ **A.** A collection of wires that transmit data and signals between components
- ○ **B.** A type of secondary storage device

- ○ **C.** A program that manages memory allocation
- ○ **D.** A cooling system for the processor

Q8 (1 mark)

How does increasing clock speed affect CPU performance?

- ○ **A.** It decreases the number of instructions that can be processed
- ○ **B.** It reduces power consumption
- ○ **C.** It increases the amount of data that can be stored
- ○ **D.** It allows more instructions to be processed per second

Q9 (1 mark)

What is an embedded system?

- ○ **A.** A computer system built into a larger device to perform specific dedicated functions
- ○ **B.** A removable component that can be added to any computer
- ○ **C.** A portable computer designed for general use
- ○ **D.** A backup system stored in the cloud

Q10 (1 mark)

Which of the following is an example of an embedded system?

- ○ **A.** A desktop computer used for office work
- ○ **B.** A laptop used for web browsing
- ○ **C.** The control system in a washing machine
- ○ **D.** A tablet used for reading ebooks

Q11 (1 mark)

What is the advantage of a CPU having multiple processor cores?

- ○ **A.** It allows multiple tasks to be processed simultaneously
- ○ **B.** It increases the storage capacity of the computer
- ○ **C.** It displays graphics more clearly
- ○ **D.** It eliminates the need for RAM

Q12 (1 mark)

How does cache memory improve CPU performance?

- ○ **A.** By permanently storing the operating system
- ○ **B.** By connecting the CPU to external storage devices
- ○ **C.** By increasing the number of processor cores
- ○ **D.** By storing frequently used data close to the CPU for faster access

Q13 (2 marks)

A program instruction needs to add two numbers stored in memory. Place the FDE cycle stages in the correct order for processing this instruction.

- ○ **A.** Execute, Decode, Fetch
- ○ **B.** Decode, Fetch, Execute
- ○ **C.** Fetch, Decode, Execute
- ○ **D.** Execute, Fetch, Decode

Q14 (2 marks)

During program execution, how do the control unit and ALU work together?

- ○ **A.** The control unit sends signals to the ALU directing it to perform specific operations, and the ALU carries out the calculations
- ○ **B.** The ALU tells the control unit what instructions to fetch
- ○ **C.** They operate independently with no communication
- ○ **D.** The ALU stores instructions while the control unit executes them

Q15 (2 marks)

A CPU is advertised as having a clock speed of 3.5 GHz. What does this specification indicate?

- ○ **A.** The CPU can store 3.5 gigabytes of data
- ○ **B.** The CPU generates 3.5 billion timing pulses per second,

allowing up to 3.5 billion instruction cycles per second

- ○ **C.** The CPU has 3.5 processor cores
- ○ **D.** The CPU weighs 3.5 grams

Q16 (2 marks)

Which characteristic best distinguishes an embedded system from a general-purpose computer?

- ○ **A.** Embedded systems are always larger than general-purpose computers
- ○ **B.** Embedded systems can run any software the user chooses
- ○ **C.** Embedded systems always require an internet connection
- ○ **D.** Embedded systems are designed for specific dedicated tasks and typically cannot be reprogrammed by the user

Q17 (2 marks)

A CPU needs to read data from a specific memory location. What roles do the different buses play in this operation?

- ○ **A.** Only the data bus is needed; addresses are not required
- ○ **B.** The address bus carries the memory location, the control bus signals a read operation, and the data bus carries the retrieved data back to the CPU
- ○ **C.** All buses carry the same information simultaneously
- ○ **D.** The buses are only used for writing, not reading

Q18 (2 marks)

A single-core CPU runs at 4 GHz. A quad-core CPU runs at 2 GHz per core. Which is likely to perform better for running multiple applications simultaneously?

- ○ **A.** The single-core CPU because 4 GHz is faster than 2 GHz
- ○ **B.** Both would perform identically in all situations
- ○ **C.** The quad-core CPU because it can process four tasks in parallel, despite each core being slower
- ○ **D.** Neither can run multiple applications

Q19 (2 marks)

A student lists the following devices: (1) Smart thermostat, (2) Gaming laptop, (3) Traffic light controller, (4) Fitness tracker. Which are embedded systems?

- ○ **A.** Smart thermostat, traffic light controller, and fitness tracker only
- ○ **B.** Only the gaming laptop
- ○ **C.** All four devices
- ○ **D.** None of them are embedded systems

Q20 (2 marks)

Why are registers essential to the fetch-decode-execute cycle?

- ○ **A.** Registers permanently store the operating system

- ○ **B.** Registers provide extremely fast temporary storage for data and instructions that the CPU is actively processing
- ○ **C.** Registers connect the computer to the internet
- ○ **D.** Registers display the current instruction to the user

Q21 (2 marks)

A CPU has 512KB of L2 cache and 8MB of L3 cache. Why are there different levels of cache rather than one large cache?

- ○ **A.** Different levels store different file types
- ○ **B.** Only one level is actually used during operation
- ○ **C.** Multiple cache levels are purely for marketing purposes
- ○ **D.** Smaller L2 cache is faster but more expensive per byte; larger L3 provides more capacity at slightly lower speed, balancing speed and size effectively

Q22 (2 marks)

What would happen if a CPU had no clock to provide timing signals?

- ○ **A.** The CPU would work faster without timing constraints
- ○ **B.** The CPU would only work for specific applications
- ○ **C.** CPU components would not be synchronised, making coordinated operation impossible and causing the processor to fail
- ○ **D.** The CPU would automatically create its own timing from solar power

Q23 (2 marks)

Why are embedded systems often more reliable than general-purpose computers for their specific tasks?

- ○ **A.** Embedded systems are optimised for their specific function, with simpler software that has fewer potential failure points than complex general-purpose systems
- ○ **B.** Embedded systems are made from more expensive materials
- ○ **C.** Embedded systems never contain any software
- ○ **D.** Embedded systems are always connected to backup power supplies

Q24 (3 marks)

Trace through the complete FDE cycle for an instruction that adds the contents of two memory locations and stores the result in a third location. Describe what happens at each stage.

- ○ **A.** The instruction executes first, then the result is decoded, and finally it is fetched from memory
- ○ **B.** Fetch: the add instruction is retrieved from memory; Decode: the control unit interprets the instruction and identifies the memory locations involved; Execute: the ALU retrieves the two values, performs the addition, and stores the result in the destination location
- ○ **C.** The entire operation happens instantaneously with no distinct stages
- ○ **D.** Only the Fetch stage is needed; Decode and Execute are not required for mathematical operations

Q25 (3 marks)

Analyse how clock speed, number of cores, and cache size each contribute to CPU performance, and explain why simply maximising one factor is not always the best approach.

- ○ **A.** Only clock speed matters; cores and cache have no effect on performance
- ○ **B.** All three factors are identical in their effect and can be substituted for each other
- ○ **C.** These factors only affect gaming and have no impact on normal computer use
- ○ **D.** Clock speed increases instructions per second per core; multiple cores enable parallel processing; larger cache reduces memory access delays. Maximising one factor has diminishing returns and trade-offs - higher clock speeds generate more heat, more cores do not help single-threaded tasks, and cache has cost and size limits

Q26 (3 marks)

A car manufacturer is designing the engine management system for a new vehicle. Explain why they would use an embedded system rather than a general-purpose computer for this purpose.

- ○ **A.** General-purpose computers are too expensive for use in cars
- ○ **B.** An embedded system provides reliable real-time control optimised for engine management, with fast response times, low power consumption, compact size, and dedicated software that starts instantly without boot delays - critical requirements that general-purpose computers cannot reliably meet
- ○ **C.** There is no difference; either would work equally well
- ○ **D.** Cars cannot contain any computer systems

Q27 (3 marks)

Explain how main memory and the CPU interact during the fetch-decode-execute cycle, including the role of buses in this communication.

- ○ **A.** Memory and CPU do not interact; they operate completely independently
- ○ **B.** Memory sends instructions to the CPU automatically without any request
- ○ **C.** The CPU places a memory address on the address bus, control signals on the control bus indicate a read operation, memory responds by placing the requested data on the data bus, and the CPU receives this data for processing; this interaction occurs repeatedly throughout the FDE cycle
- ○ **D.** The CPU contains all necessary data internally and never needs to access memory

Q28 (3 marks)

The ALU is described as performing 'arithmetic and logic' operations. Analyse these two categories of operations and provide examples of when each would be used during program execution.

- ○ **A.** Arithmetic and logic are different names for the same type of operation
- ○ **B.** The ALU only performs arithmetic; logic operations are handled elsewhere
- ○ **C.** Logic operations are only used for connecting to the internet
- ○ **D.** Arithmetic operations include calculations like addition, subtraction, multiplication, and division used when computing values; logic operations include comparisons (equal, greater than, less than) and Boolean operations (AND, OR, NOT) used in decision making and conditional statements

Q29 (3 marks)

Compare embedded systems and general-purpose computers in terms of: (1) flexibility, (2) user interaction, and (3) typical applications.

- ○ **A.** There are no meaningful differences between embedded and general-purpose systems
- ○ **B.** Embedded systems are more flexible because they can be reprogrammed easily
- ○ **C.** Embedded systems are specialised for fixed tasks with minimal or no user interface, found in appliances and devices; general-purpose computers are flexible, run user-chosen software with full user interfaces, and serve varied applications from office work to gaming
- ○ **D.** General-purpose computers can only perform one task while embedded systems handle multiple tasks

Q30 (3 marks)

A student is choosing between two CPUs: CPU A has 6 cores at 3.0 GHz with 6MB cache; CPU B has 4 cores at 4.5 GHz with 12MB cache. Analyse which might perform better for: (1) video editing with multi-threaded software, (2) playing a single-threaded game, (3) general multitasking.

- ○ **A.** CPU A is better for all tasks because it has more cores
- ○ **B.** CPU B is better for all tasks because it has higher clock speed
- ○ **C.** Video editing likely favours CPU A (more cores for parallel processing); single-threaded gaming favours CPU B (higher clock speed); multitasking depends on workload but both would perform adequately with CPU A better for many simultaneous applications and CPU B better if applications benefit from speed and cache
- ○ **D.** Both CPUs would perform identically in all scenarios

Total: **55** marks • Time: **30 min**

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